

AI-Powered Job Recommender

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Semantic Matching AI Revolution

The Context Problem:

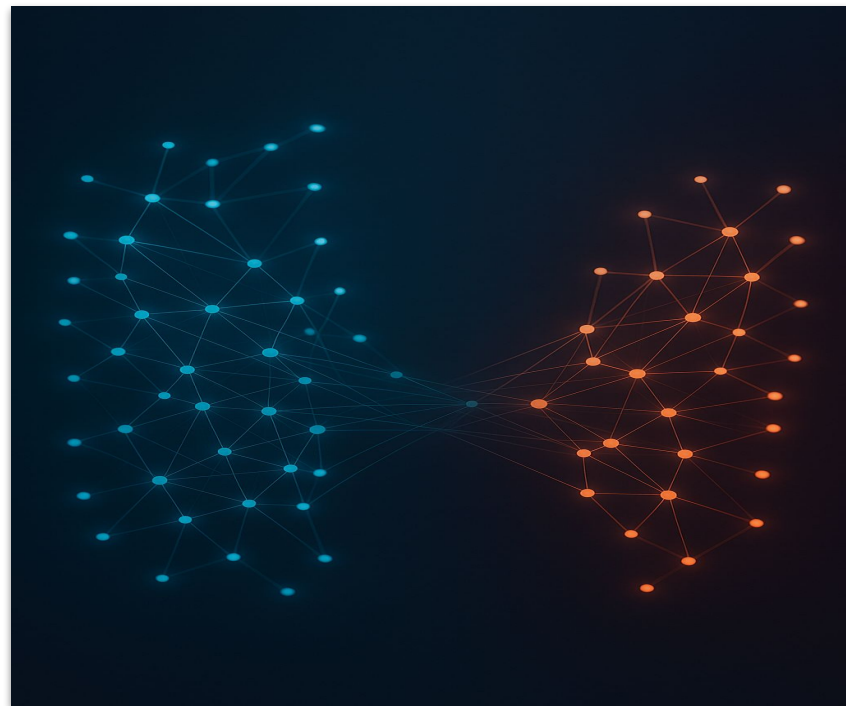
- Keyword matching misses 40% of relevant opportunities

Our Solution: SBERT Hybrid Model

- Two-stage architecture: Fast filtering + Semantic understanding
- Combines TF-IDF efficiency with transformer intelligence
- Understands context, not just keywords

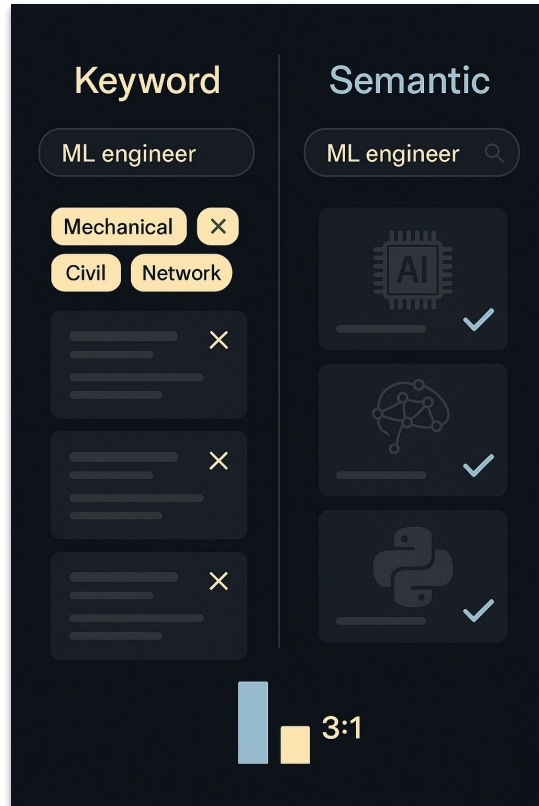
Headline Performance:

- **95.9%** Ranking Accuracy (nDCG@5)
- **73.1%** Relevant Recommendations
- **114ms** Real-time Response



Where Traditional Matching Fails

- **Semantic Limitations:**
 - Keywords don't capture contextual meaning or skill transferability
- **Information Asymmetry:**
 - Poor filtration leads to information overload for both job seekers and recruiters
- **Transparency Gap:**
 - Existing systems lack interpretability



Data Foundation

Resume Dataset:

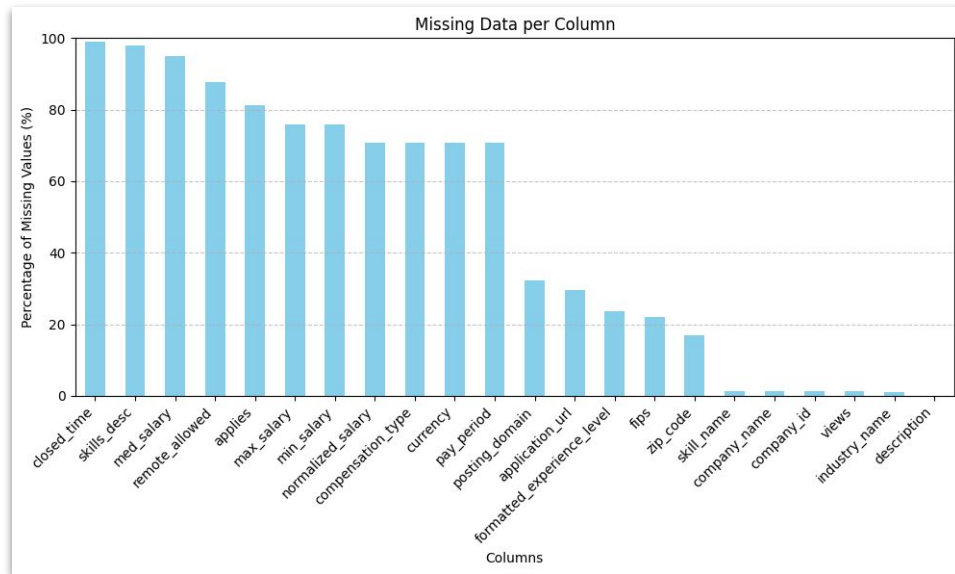
- **2,484 Professional Resumes**
- **24 Job Categories:** HR, Engineering, Sales, Healthcare, Technology, Business Development, etc.
- Ground-truth category labels for evaluation
- Structured text with career progression data

LinkedIn Job Dataset:

- **123,849 Raw Postings** (2023–2024)
- **105,000 Quality Listings** after filtering
- Current market conditions representation
- Wide industry coverage across sectors

Data Quality Reality:

- **60–80% Missing Metadata:** No structured skills, salary, or requirement fields
- **Category Imbalance:** Some categories 40x overrepresented
- **Format Chaos:** HTML artifacts, encoding issues, inconsistent structures
- **Length Variance:** 50 to 5000+ words per document



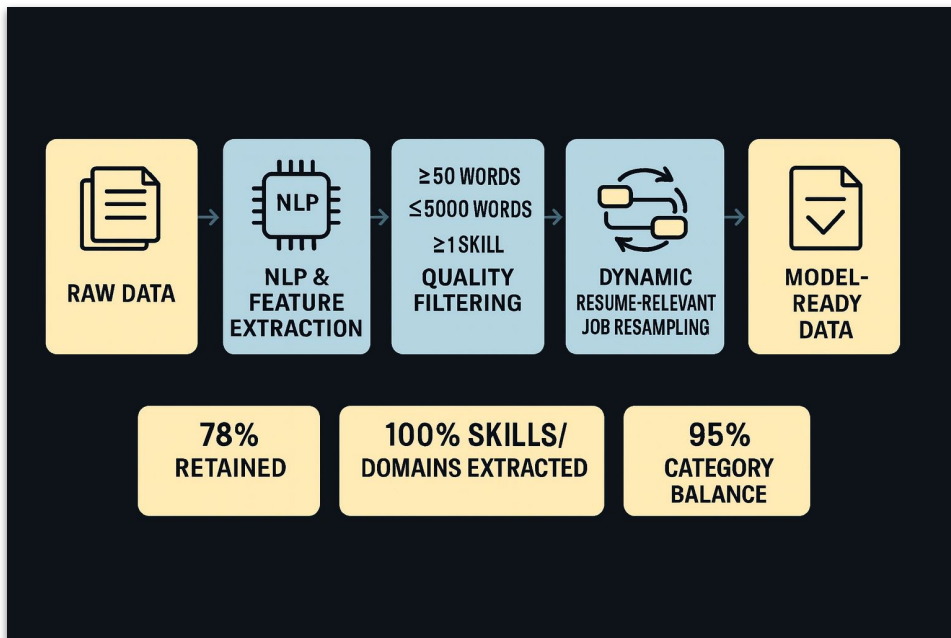
Data Preprocessing + Feature Engineering

Preprocessing Pipeline:

- Text Normalization & Quality Filtering**
 - Removed URLs, emails, special characters
 - Applied hard thresholds: 50-5000 words, minimum 1 skill
 - Quantile filtering (2.5th-97.5th percentiles) per category
- SpaCy NLP Processing**
 - Tokenization & lemmatization with `en_core_web_lg`
 - Custom skill extraction (`jz_skill_patterns.jsonl`)
 - Named entity recognition for domains
- Adaptive Quality Filtering**
 - Category-specific thresholds for text complexity
 - Lexical diversity requirements per professional field
 - Preserved professional terminology
- Domain-Aware Resampling**
 - Mapped resumes to relevant job domains
 - Created career bridge connections
 - Balanced representation across categories

Performance Metrics:

- 78% document retention (from 123K → 105K jobs)
- 95% category balance maintained
- 100% skill/domain extraction success
- 300+ documents/second processing speed



Exploratory Insights: Skills & Complexity

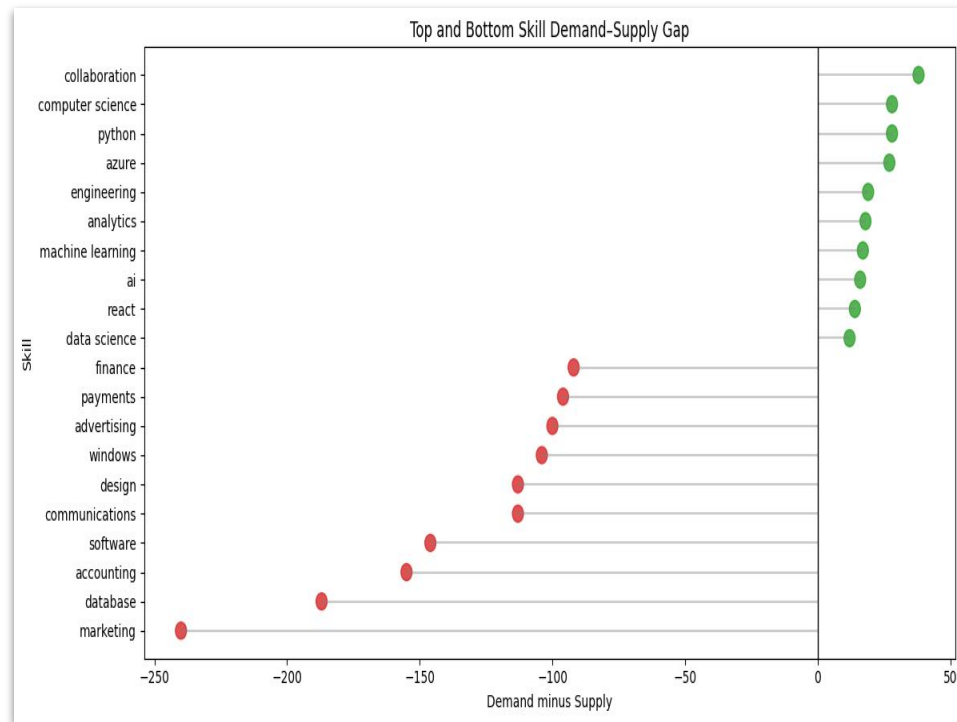
Text Complexity Patterns:

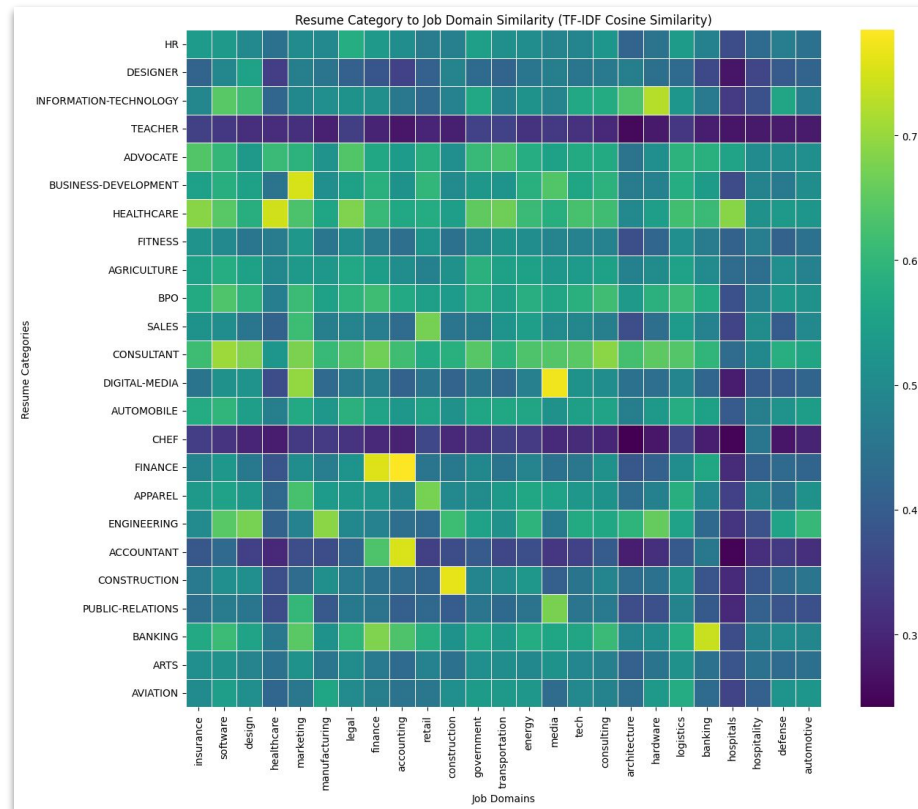
- Lexical diversity ↔ Skill count correlation: $r = 0.42$
- Executive resumes: 72% lexical diversity
- Technical roles: 58% lexical diversity
- Text length ≠ Professional expertise indicator

Critical Skill Gaps Discovered:

- **Cloud Computing:** 3:1 demand-to-supply ratio
- **Data Science:** 250% undersupply gap
- **Machine Learning/AI:** High demand, low candidate availability
- **Traditional Business Skills:** Near equilibrium (1:1)

Key Finding: Professional categories have distinct "linguistic fingerprints" that reveal career patterns





LLM Evaluation Rubric (0-5)

Human-like relevance scoring for resume ↔ job matches

Evaluation Criteria

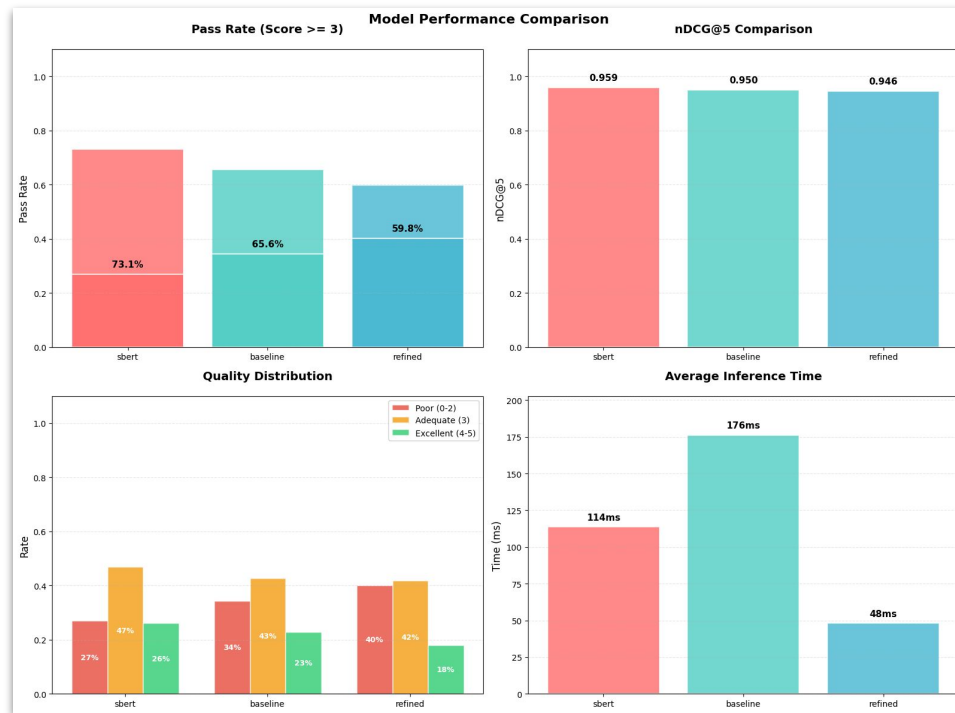
- 1) Skills alignment with job requirements
- 2) Experience level appropriateness
- 3) Industry / domain relevance

Scoring: JSON {"score": X, "justification": "..."}

Score	Label	Definition
5	ROCKSTAR	Perfect candidate exceeding all qualifications
4	SOLID PASS	Strong candidate for immediate interview
3	PASS	Qualified candidate worth considering
2	INSUFFICIENT	Relevant background, missing key requirements
1	WEAK	Limited correlation between resume and job
0	IRRELEVANT	Completely unrelated match

Results & Performance

- **Baseline TF-IDF:**
 - Standard cosine similarity over full corpus
- **Refined TF-IDF:**
 - Improved preprocessing of corpus
 - Improved inference time
 - Decreased recommendation relevancy
- **SBERT Hybrid:**
 - Two-Stage Pipeline
 - TF-IDF Candidate Generation followed by semantic re-ranking and domain boosting
- **Hybrid Improvement:**
 - +0.9 % ranking accuracy
 - +13.3 % relevance over refined model
 - 44 % more high-quality recommendations.



Model	nDCG@5	Pass Rate ($\geq 3/5$)	Quality Rate ($\geq 4/5$)	Inference Time
SBERT Hybrid	0.959	73.1%	26.1%	114ms
Baseline TF-IDF	0.950	65.6%	22.9%	176ms
Refined TF-IDF	0.946	59.8%	18.1%	48ms

Lessons & Impact

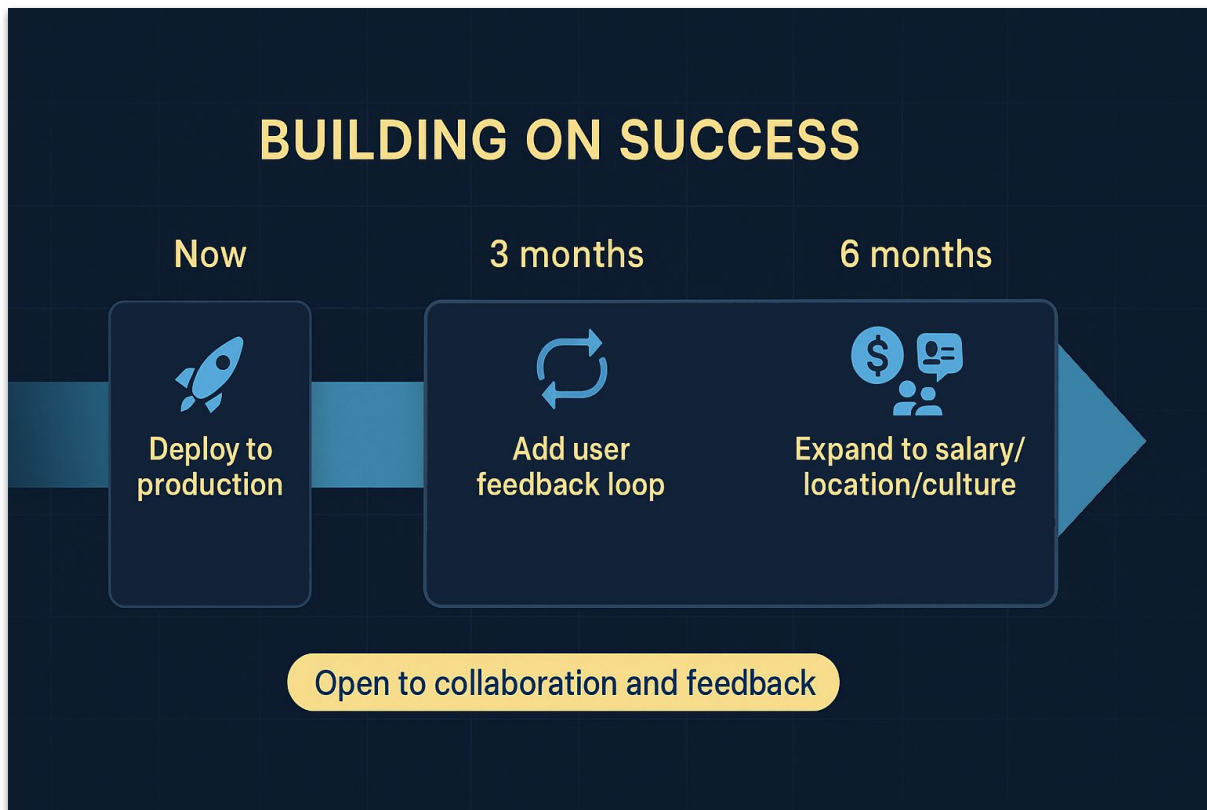
Technical Achievements:

- ✓ Semantic > Keywords
- ✓ Two-stage architecture works
- ✓ AI can evaluate AI

Business Impact:

- 23% better matches
- 3/4 relevant recommendations
- Real-time performance

Next Steps



Thank You!
